

What does a decoder believe?

exp094 · 22 August 2026 · Complete

Introduction

A decoder converts neural activity into a class decision. Here the learned output projection remains frozen while two later choices change: z decides what temporal activity counts as evidence, and p decides how that evidence is displayed as relative confidence.

One preselected MNIST digit and one Poisson spike tensor were replayed through the canonical seed-42 COBA and PING final-epoch checkpoints represented in the latest gold-star publication view. These are the full-MNIST reference networks, not exp022's 10%-MNIST activity-frontier (off) condition. Their trained decoder is **mem-mean**: a non-spiking output membrane whose temporal mean supplies the class logits. To compare spike-based alternatives without retraining, the same frozen hidden activity and learned output weights also drive a counterfactual output-LIF **spike-count** decoder at inference time. Every spike-based z reads that one shared intervention trajectory, and every p reads the same cumulative counts. The comparison therefore explains decoder choices; it does not estimate the accuracy each alternative would achieve if trained end to end. Each subsection pairs a blue-grey qualitative mechanism diagram with a structurally matched red-black measured mirror.

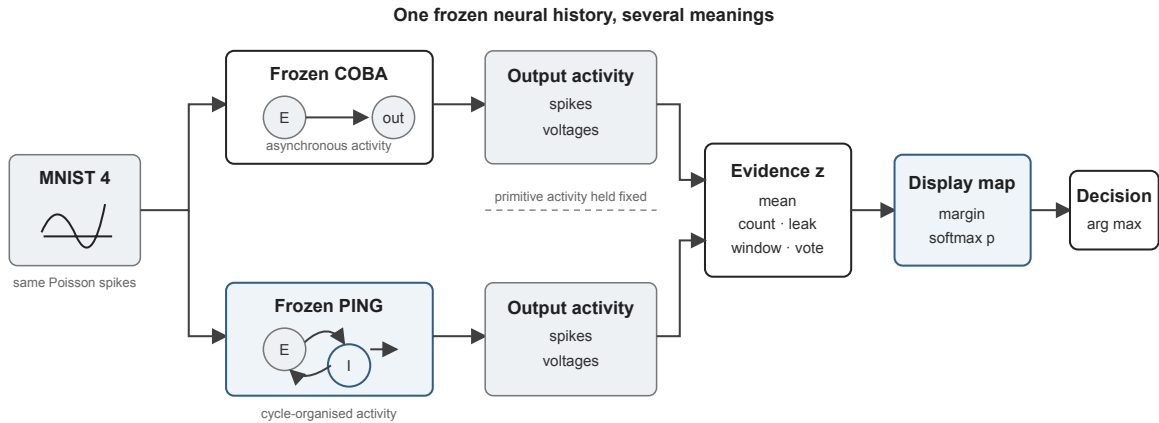


Figure 1: Decoder decomposition. Frozen COBA and PING networks produce output spikes and voltages. A temporal evidence function z compresses that activity, an output mapping p displays the resulting competition, and the decision selects the largest final score.

Temporal evidence functions z

Each subsection changes only z . In the measured plots, red denotes the true class, black its strongest competitor, and time runs horizontally; the diagrams use blue and grey to remain visibly distinct. Mean voltage uses the native non-spiking readout. The four spike-based alternatives use the single counterfactual output-LIF trajectory driven by the same frozen hidden spikes and output weights.

Mean output voltage

$$z_c^{\text{mean}(t)} = \frac{1}{t+1} \sum_{\tau=0}^t v_c^{\text{pre}(\tau)}.$$

Here c identifies a class, t is the current timestep, τ indexes earlier timesteps, and $v_c^{\text{pre}(\tau)}$ is pre-reset output voltage. Mean voltage retains subthreshold evidence but removes temporal order. It is the decoder used to train the canonical exp022 COBA and PING checkpoints.

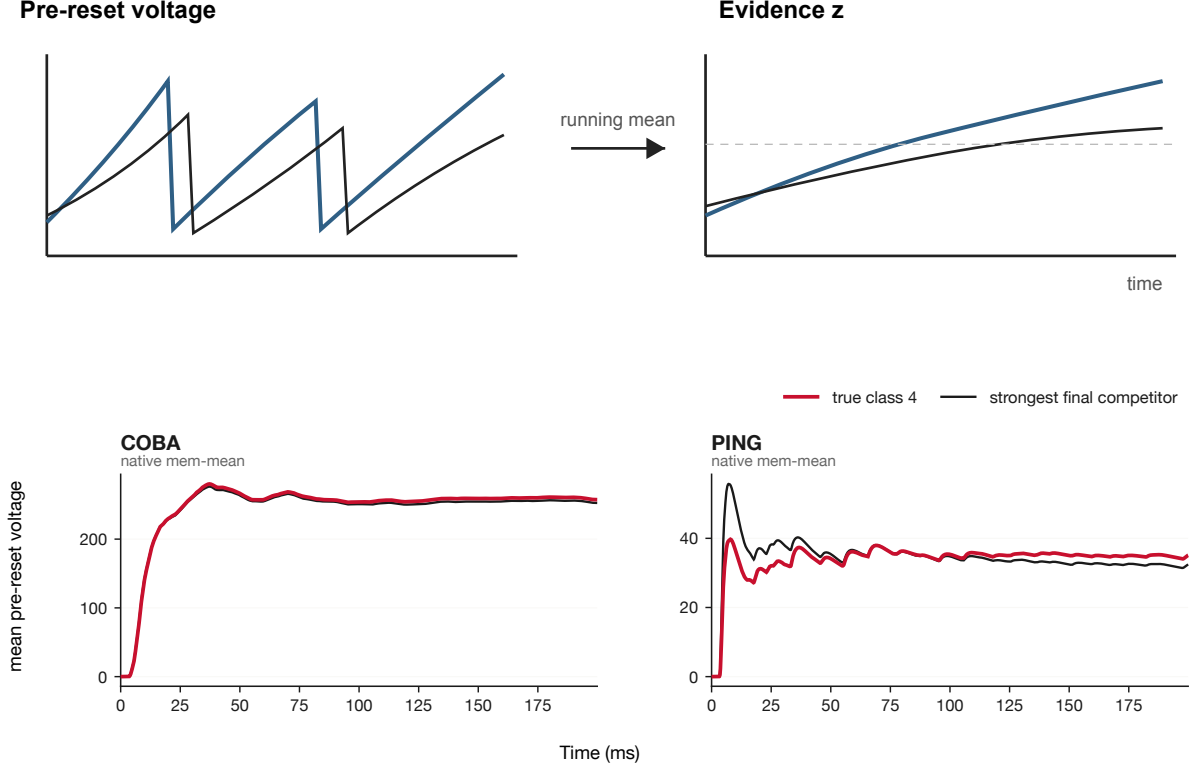


Figure 2: Mean-voltage decoder. The diagram shows pre-reset voltages entering a running temporal mean; the measured mirror shows the resulting frozen COBA and PING trajectories.

The native decoder selects the true class in both networks. COBA settles rapidly, whereas PING retains a visible early competition before the true-class mean becomes consistently larger.

Cumulative spike count

$$z_c^{\text{cum}(t)} = \sum_{\tau=0}^t s_{c(\tau)}.$$

Here $s_{c(\tau)}$ is one when output neuron c spikes at timestep τ and zero otherwise. Every spike remains evidence until reset. Timing and order disappear; only accumulated count survives.

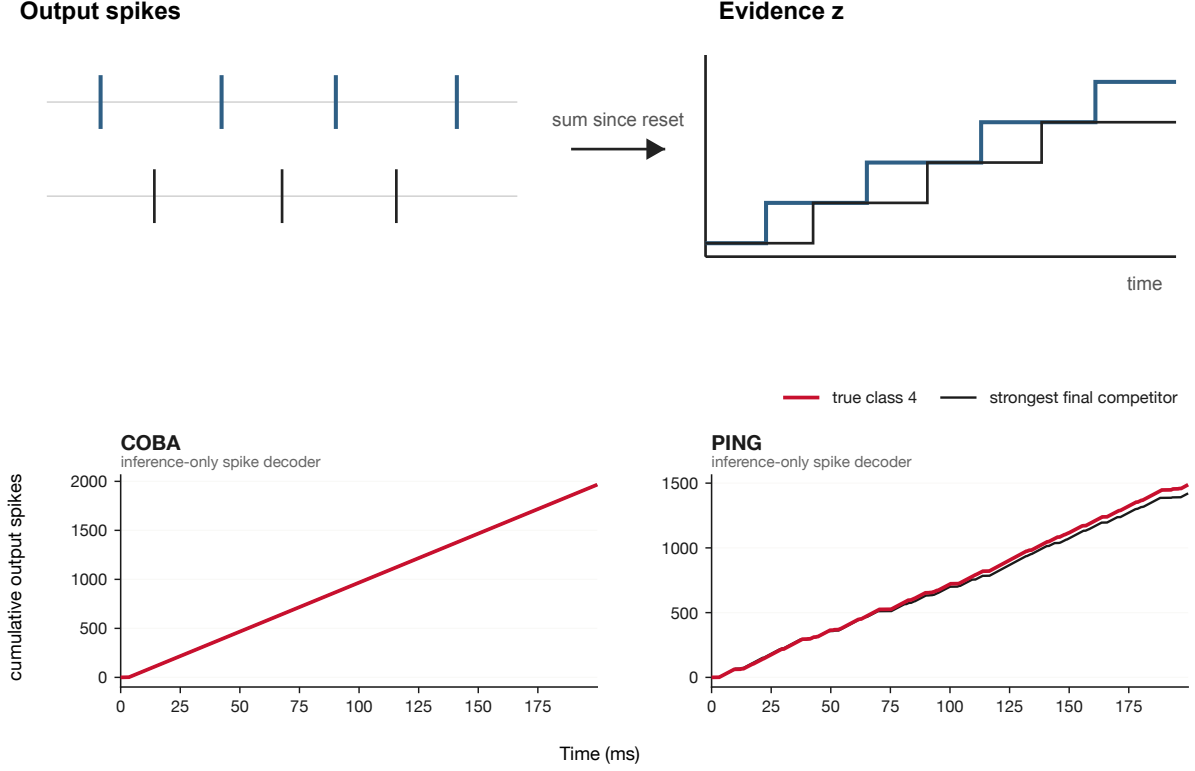


Figure 3: Cumulative-count decoder. Each output spike increments one class trace and no evidence decays; the staircase exposes early leads and later reversals.

The counterfactual spike decoder changes the outcome. PING eventually gives the true class the largest cumulative count, while COBA does not. Frozen weights therefore do not make a new temporal evidence rule equivalent to the rule used during training.

Leaky spike count

$$z_c^{\text{leak}(t)} = \lambda z_c^{\text{leak}(t-1)} + s_{c(t)}, \quad 0 < \lambda < 1.$$

Here λ is the fraction of previous evidence retained for one timestep. Spikes add evidence; between spikes it decays. The decoder can follow changing input, but its meaning depends on the chosen memory timescale.

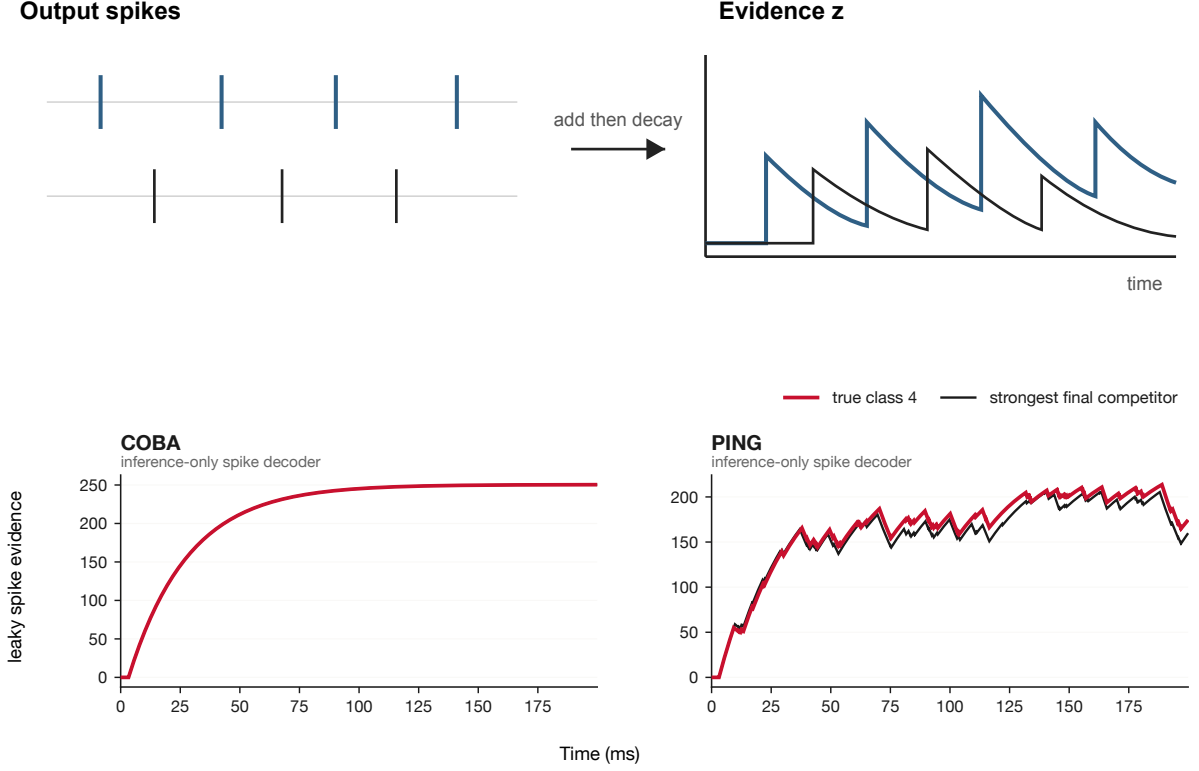


Figure 4: Leaky-count decoder. Spikes create upward steps and evidence decays smoothly between them, using the same leak factor for COBA and PING.

Leakage exposes recent differences in PING rather than preserving its entire history. COBA’s much denser counterfactual output keeps both displayed classes near a steady ceiling, so forgetting adds little useful separation.

Sliding-window spike count

$$z_c^{\text{win}(t)} = \sum_{\tau=\max(0, t-W+1)}^t s_c(\tau).$$

Here W is the number of timesteps in the window. A spike contributes while it remains inside the window and disappears at the trailing boundary. Memory is finite and explicit rather than gradual.

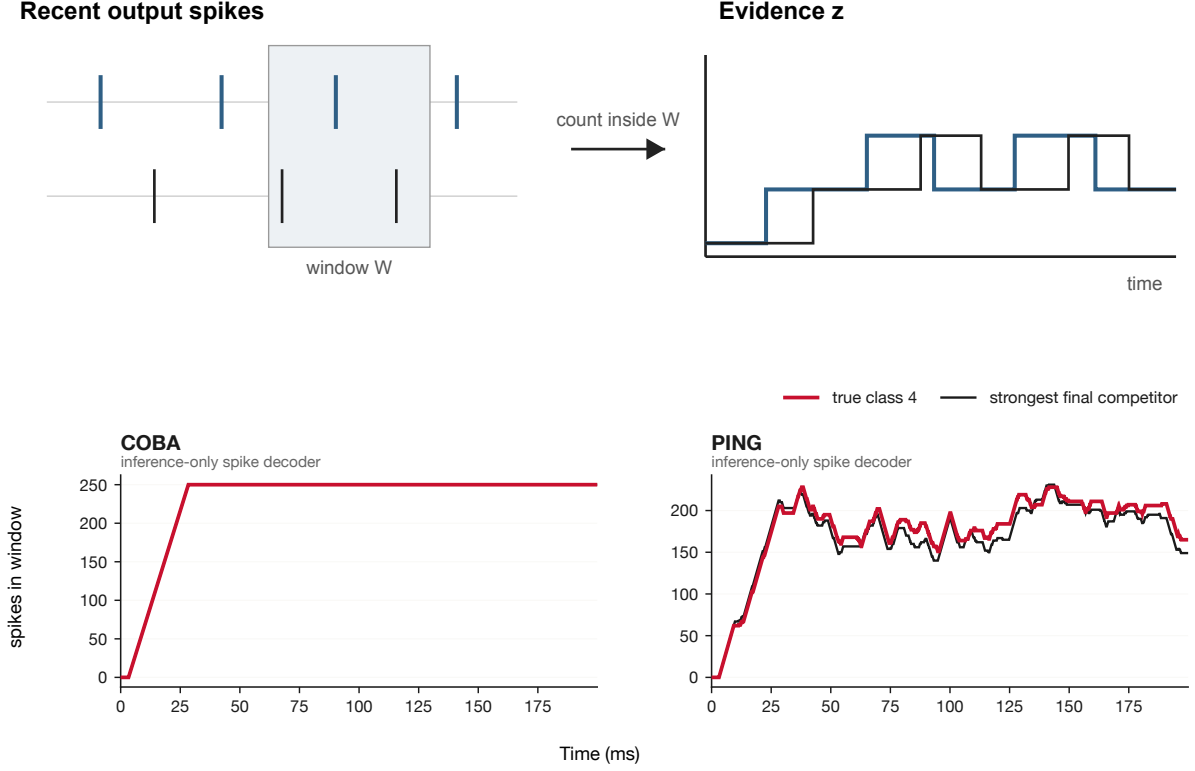


Figure 5: Sliding-window decoder. Evidence persists for 25 ms, then disappears abruptly when its spike leaves the window.

The finite window makes PING’s changing local competition explicit. COBA again remains saturated: nearly every window is full, showing that a memory rule cannot rescue an output representation whose event rate is poorly matched to it.

Cycle or matched-bin voting

$$z_c^{\text{vote}(K)} = \sum_{k=1}^K 1_{c=\arg \max_j n_{j,k}}.$$

Here K is the number of completed bins, k identifies one bin, j indexes classes, and $n_{j,k}$ is class j ’s spike count inside bin k . The subscripted one is one when its condition is true and zero otherwise. PING bins follow measured cycles; COBA uses equal-duration controls. Each bin contributes one vote, discarding within-bin burst size.

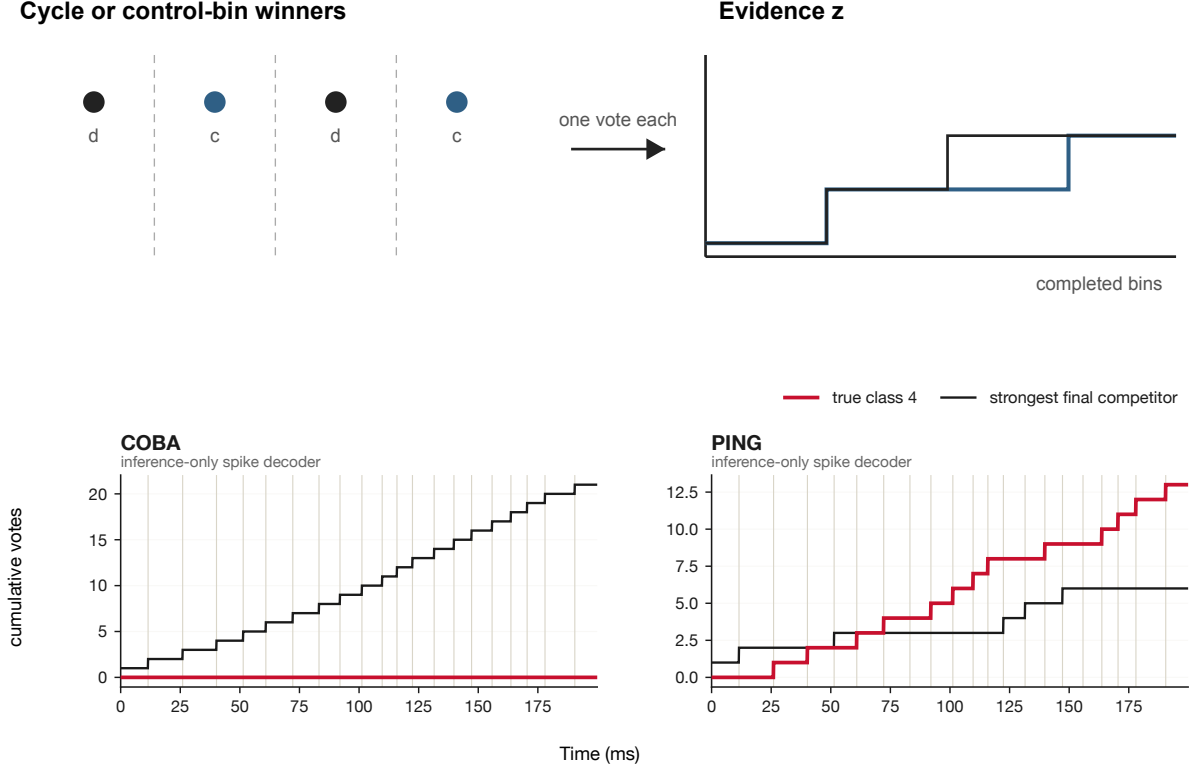


Figure 6: Cycle or matched-bin voting. Each interval names one winner before adding one vote. PING votes follow detected cycles and COBA votes use duration-matched controls.

PING’s true class wins more detected cycles and pulls away in cumulative votes. The matched COBA bins consistently favour another class. Cycle voting is therefore meaningful here because PING supplies distinct rhythmic evidence packets, not because voting itself guarantees a better decision.

Output mappings p

Controlled comparison. All three p mappings receive the identical ten-class cumulative-count vector

$$\mathbf{z}^{\text{cum}(t)} = \left(z_0^{\text{cum}(t)}, \dots, z_9^{\text{cum}(t)} \right).$$

The counterfactual output-LIF trajectory is generated once from the shared frozen hidden drive, and $\mathbf{z}^{\text{cum}(t)}$ is calculated once. It is then passed unchanged to ordinary softmax, temperature-softened softmax, and independent sigmoid. We do not select or recompute a different $\mathbf{z}$ for any p . Therefore, every difference between these three displays is caused by p alone.

Here $\mathbf{z}^{\text{cum}(t)}$ is the ten-element cumulative-count vector at time t , and its entries $z_0^{\text{cum}(t)}$ through $z_9^{\text{cum}(t)}$ correspond to MNIST classes zero through nine.

Ordinary softmax

Fixed input to p : the shared cumulative-count vector $\mathbf{z}^{\text{cum}(t)}$.

$$p_{c(t)} = \frac{\exp(z_c^{\text{cum}(t)})}{\sum_j \exp(z_j^{\text{cum}(t)})}.$$

Here $p_{c(t)}$ is class c 's displayed share, $\exp$ is the exponential function, and j ranges over all ten classes. Scores sum to one and preserve the winner. Count differences are exponentially sharpened.

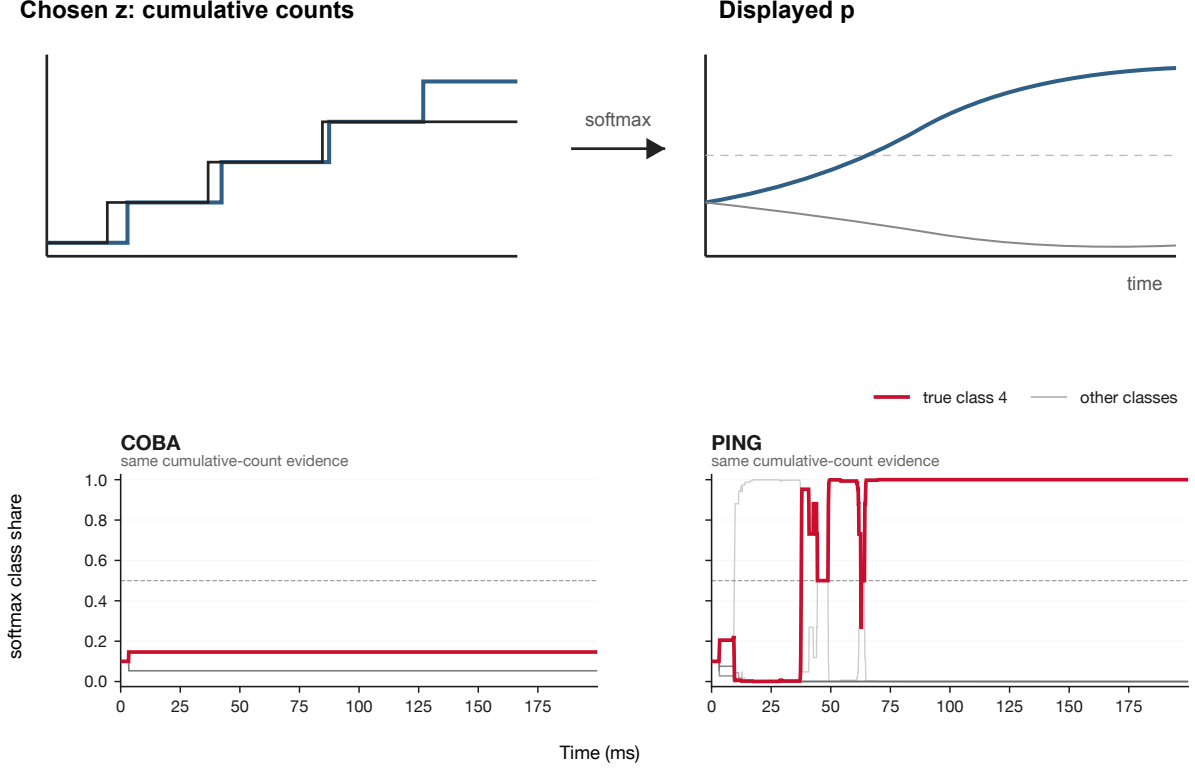


Figure 7: Ordinary softmax applied to cumulative spike counts. The measured mirror shows how an accumulating count margin becomes apparent confidence.

Softmax hides absolute activity and displays only count margins. PING's early leader changes produce abrupt transfers of displayed share before the true class dominates; COBA's different final winner remains unchanged by the mapping.

Temperature-softened softmax

Fixed input to p : the same $z^{\text{cum}(t)}$ used by ordinary softmax.

$$p_{c(t;T)} = \frac{\exp\left(\frac{z_c^{\text{cum}(t)}}{T}\right)}{\sum_j \exp\left(\frac{z_j^{\text{cum}(t)}}{T}\right)}, \quad T > 1.$$

Here T is temperature. Dividing all counts by the same positive T preserves the winner but reduces visual sharpening. This separates accumulated evidence from its apparent decisiveness.

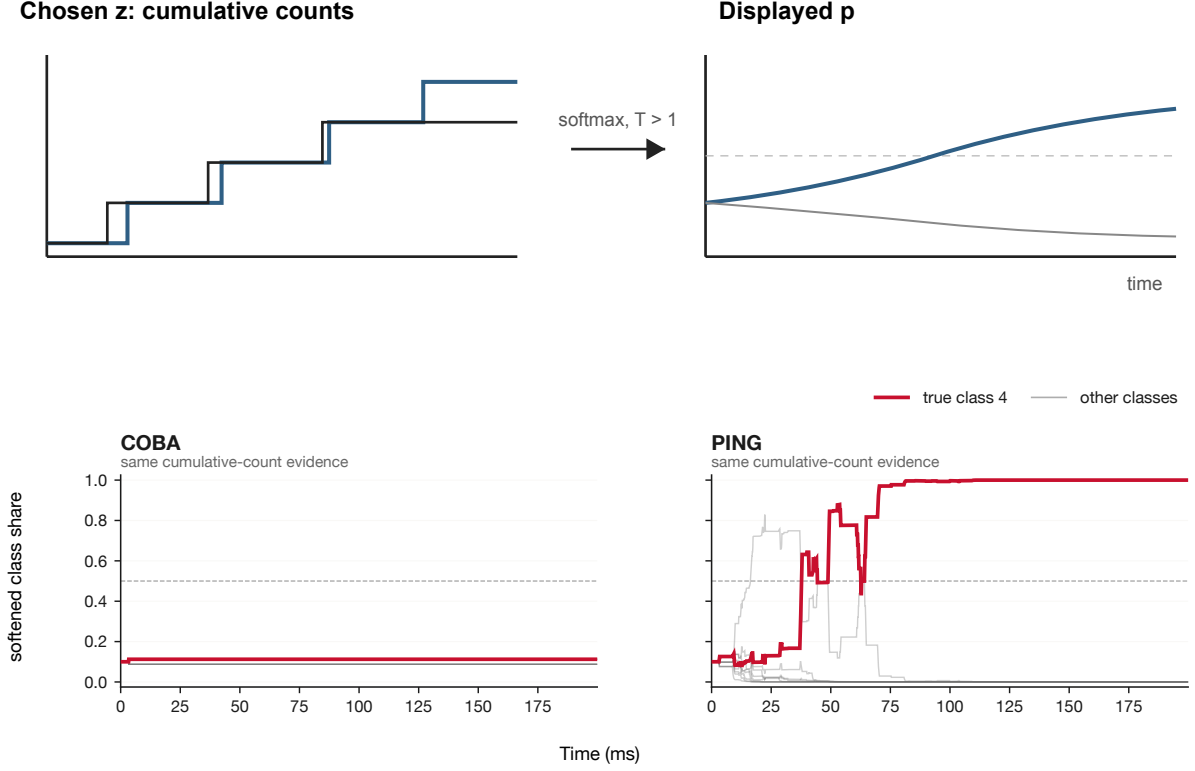


Figure 8: Temperature-softened softmax applied to the same counts. It differs from ordinary softmax only in temperature, isolating the effect of p .

Temperature preserves every leader change and the final winner while making the competing PING classes easier to see. It changes how decisive the same evidence appears, not which evidence was accumulated.

Independent sigmoid

Fixed input to p : the same $z^{\text{cum}(t)}$ used by both softmax mappings.

$$p_c^{\text{sig}(t)} = \frac{1}{1 + \exp(-z_c^{\text{cum}(t)})}.$$

Here $p_c^{\text{sig}(t)}$ is an independent bounded score. One class does not suppress the others and the ten scores need not sum to one. Since cumulative counts are non-negative and grow, several classes may approach one together. This exposes why independent sigmoid is poorly matched to mutually exclusive MNIST labels when applied directly to raw counts.

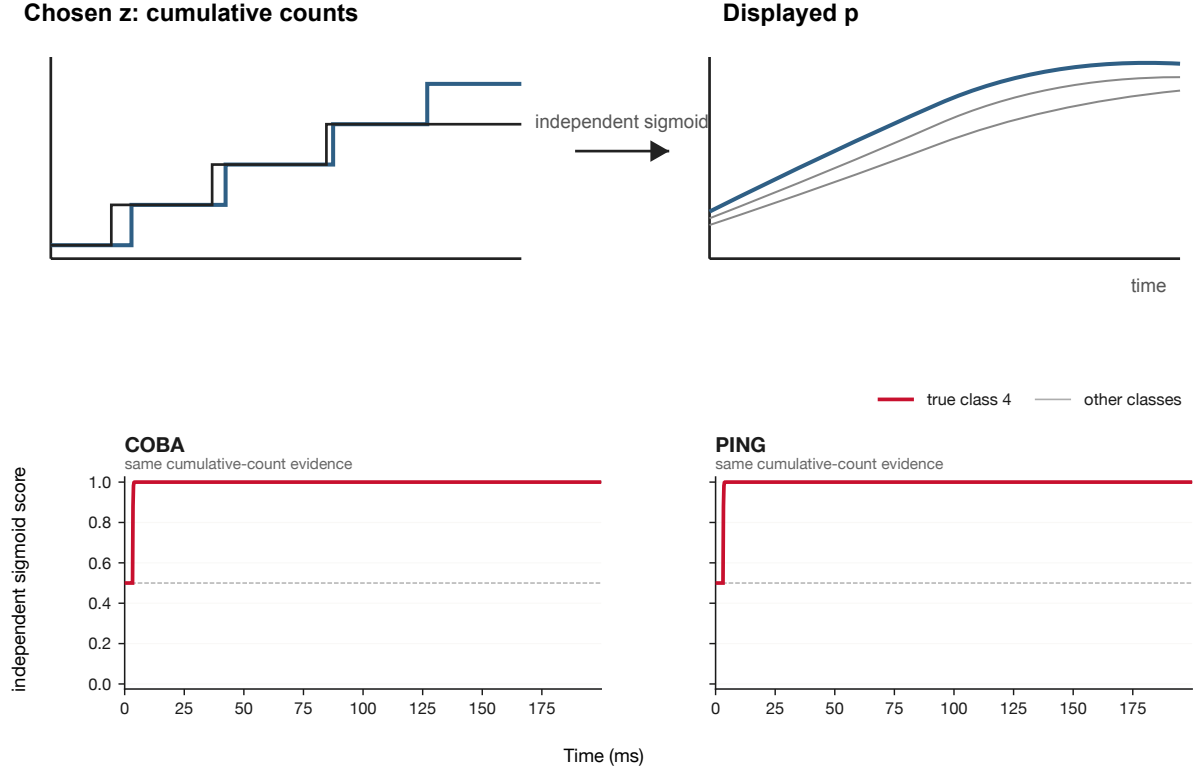


Figure 9: Independent sigmoid applied to cumulative spike counts. It retains the same z used by both softmax variants and reveals what is lost when classes no longer compete.

Because every cumulative count grows, all ten sigmoid scores saturate near one almost immediately. The mapping preserves neither useful relative separation nor a meaningful class decision; its failure follows from applying an independent bounded transform to non-negative accumulating counts.

Screening check across 100 images

The single digit above suggests that PING exposes class information through spikes more robustly than COBA, but one example cannot establish that pattern. A fast screening run therefore selected the first ten official-test images from each MNIST class without inspecting their outcomes. The same fixed Poisson encoding was presented to both frozen networks. Inputs were processed in batches, and every alternative decoder reused its recorded activity. This 100-image convenience sample completed locally in under one minute. It is a check on whether the observation immediately collapses, not an estimate of test-set accuracy.

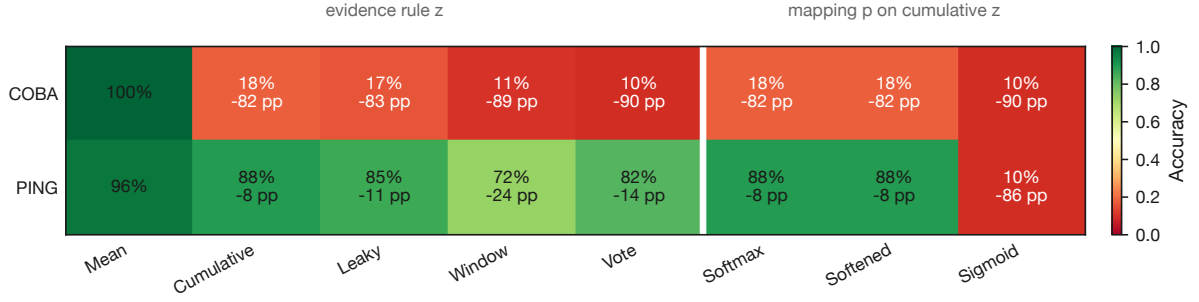


Figure 10: Decoder transfer screen. Each cell gives accuracy and, below it, the change from that network’s native mean-voltage decoder. The divider separates temporal evidence choices from mappings of the shared cumulative-count evidence.

The contrast survives the screen. COBA’s native decoder classifies all 100 images correctly, but its spike-based evidence rules retain only 10–18%. PING’s native decoder reaches 96%; cumulative count retains 88%, leaky count 85%, cycle voting 82%, and the 25 ms window 72%. This does not show that PING is decoder-independent. It shows that its spike events preserve substantially more of the trained decision than COBA’s counterfactual output spikes do.

Ordinary and softened softmax exactly match cumulative-count accuracy because both preserve its winning class. Independent sigmoid falls to 10% for both networks: the accumulating non-negative counts saturate together, and numerical ties default to class zero. Its apparent winner is therefore not a meaningful decoded decision.

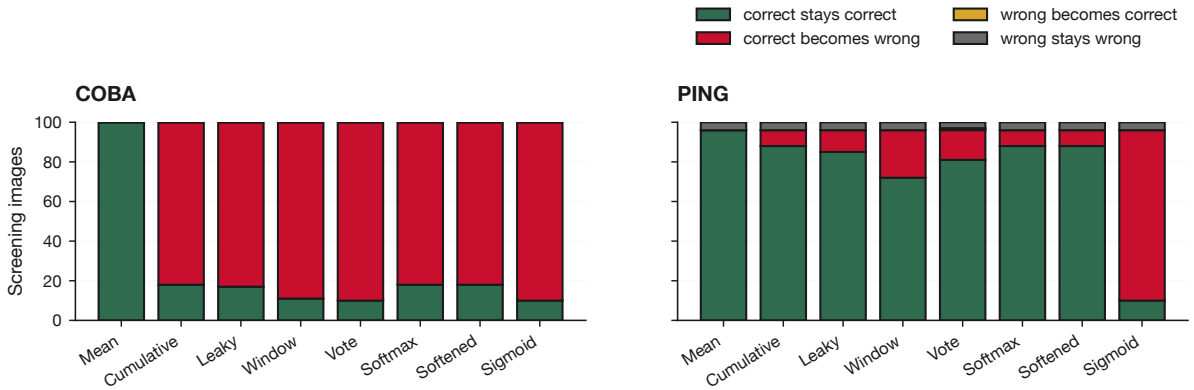


Figure 11: Decision transitions relative to native decoding. Green is preserved correctness; red is a native correct decision broken by the alternative. Yellow denotes a repair and grey a prediction that remains wrong.

COBA’s alternatives break 82–90 decisions that its native decoder gets right. PING loses 8 decisions under cumulative count, 11 under leaky count, 24 under the finite window, and 15 under cycle voting. Cycle voting also repairs one of PING’s four native errors. The distinction is therefore not merely a shift in aggregate accuracy: it is a large difference in how often changing the decoder destroys an already correct decision.

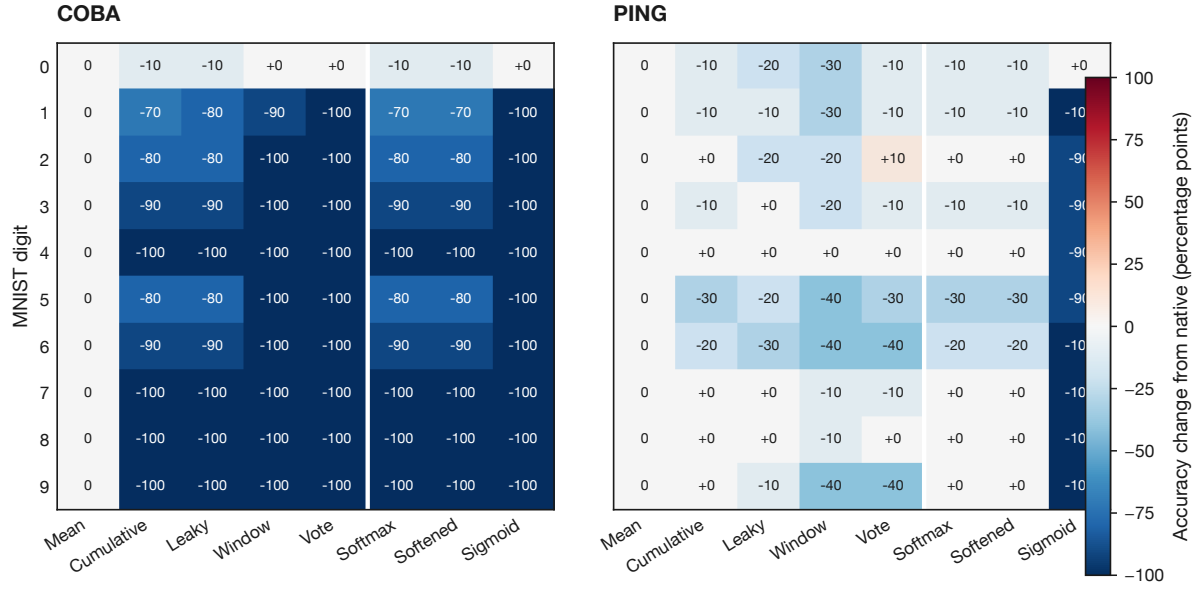


Figure 12: Class-specific change from native accuracy, in percentage points. The screen is balanced at ten images per digit, so each image changes its class cell by ten points.

COBA’s loss spans almost every digit rather than reproducing only the original class-4 failure. Its spike decoders disproportionately return class zero, explaining why digit zero is the lone robust row. PING’s losses are smaller and distributed across classes, with cumulative decoding weakest here for digits five and six. A full test-set, multi-seed evaluation is still required before treating decoder robustness as an architectural property.